Server Management

Core Server

Why use core?

- The Server Core option is a minimal installation option that is available when you are deploying the Standard or Datacenter edition of Windows Server.
- Server Core includes most but not all server roles.
- Server Core has a smaller footprint, and therefore a smaller attack surface due to a smaller code base.
- Server Core installation eliminates any services and other features that are not essential for the support of certain commonly used server roles.

https://learn.microsoft.com/en-us/windows-server/administration/server-core/what-is-server-core

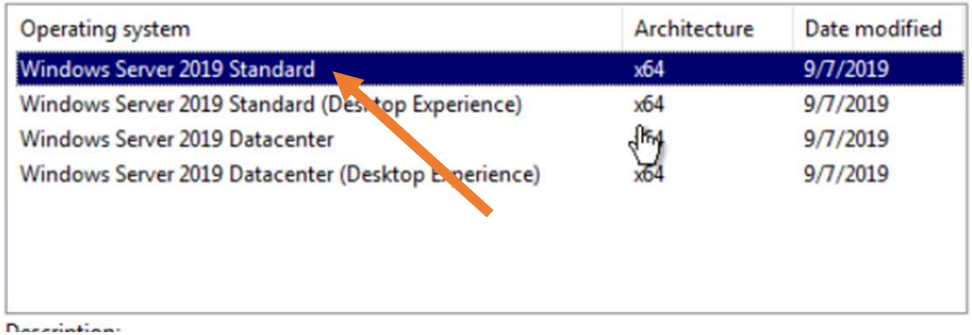
What is included with core?

- Server Core installation eliminates any services and other features that are not essential for the support of certain commonly used server roles.
- See the link below for a list of services, roles and features that differ between core and desktop experience installation.
- Services
  https://learn.microsoft.com/en-us/windows-server/administration/server-core/what-is-server-core#feature-on-demand---available-features-in-feature-on-demand-package-1809
- Roles/Features
  https://learn.microsoft.com/en-us/windows-server/administration/server-core/server-core-roles-and-services

https://learn.microsoft.com/en-us/windows-server/administration/server-core/what-is-server-core

Installing Core

- When you install Windows Server using Windows PE, you can choose between Server Core or Server with Desktop Experience install options.
  - By ***not*** installing Desktop Experience, you are creating a Core server.



Operating system	Architecture	Date modified
Windows Server 2019 Standard	x64	9/7/2019
Windows Server 2019 Standard (Desktop Experience)	x64	9/7/2019
Windows Server 2019 Datacenter	x64	9/7/2019
Windows Server 2019 Datacenter (Desktop Experience)	x64	9/7/2019

Administering Core

(Initial Configuration)

- All versions of Core (before Server 2022) will start the CMD automatically. From the CMD Windows you can launch `SCONFIG`, Powershell, and any other command normally found in Windows Server.
- The Server Configuration tool (`SConfig`) is the primary way to configure and manage common aspects of a Core Server installation.
- `SConfig` is a command line, menu-driven application
- You can use `SConfig` in the following scenarios:
  - you have signed in locally
  - you are connected with Remote desktop.
- `SConfig` cannot be used in a remote PowerShell session.

```
================================================================================
                    Welcome to Windows Server 2022 Datacenter
================================================================================

  1)  Domain/workgroup:                   Workgroup: WORKGROUP
  2)  Computer name:                      WIN-VG2A69FLC33
  3)  Add local administrator
  4)  Remote management:                  Enabled

  5)  Update setting:                     Download only
  6)  Install updates
  7)  Remote desktop:                     Disabled

  8)  Network settings
  9)  Date and time
  10) Telemetry setting:                  Required
  11) Windows activation

  12) Log off user
  13) Restart server
  14) Shut down server
  15) Exit to command line (PowerShell)

Enter number to select an option: _
```

Administering Core

(Initial Configuration)

- From `sConfig`, you can set the following configurations:
  - Domain/WorkGroup membership
  - Hostname
  - RDP configuration
  - Windows Update Settings
  - Network Settings
  - Date and time
  - Reboot and shutdown options
  - Windows Activation
  - Telemetry

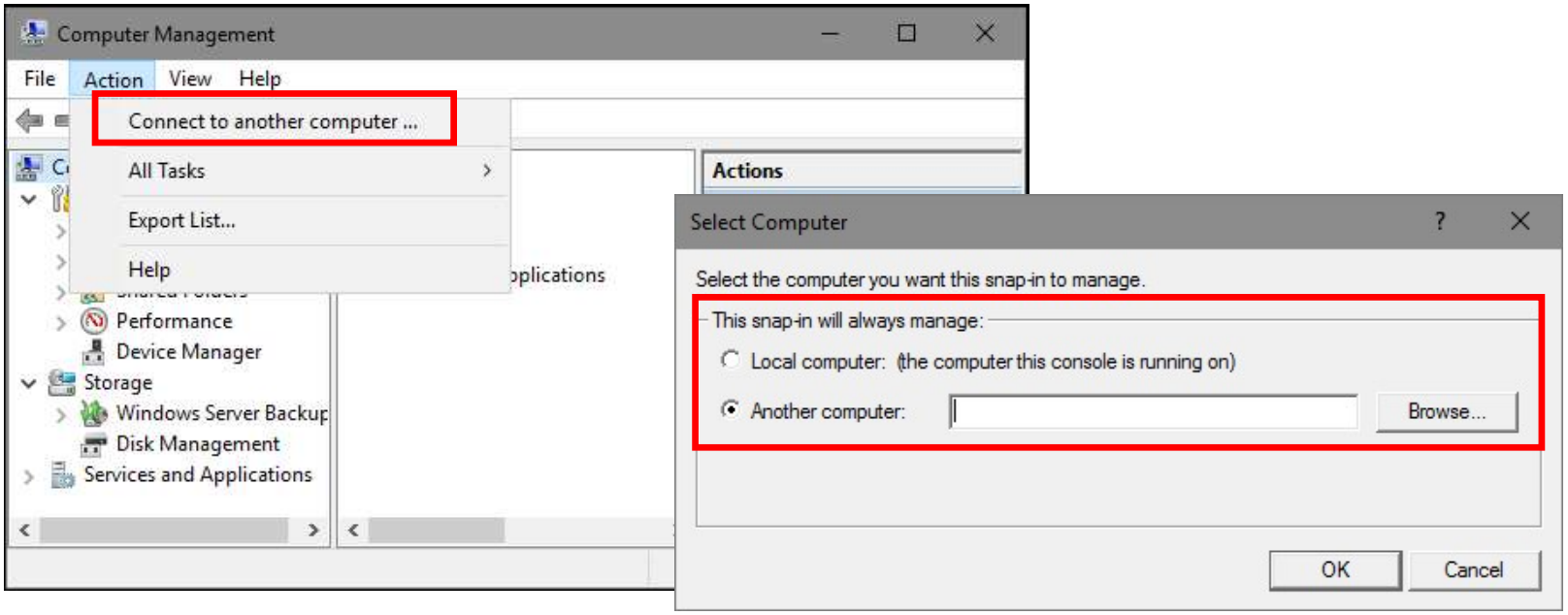
Remote Administration

Remote Administration using the MMC

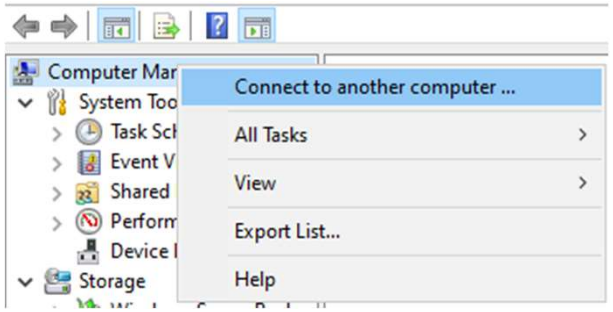
- The various pre-configured MMC consoles (compmgmt.msc, services.msc, etc) included in Windows are all configured to manage resources on the local system. However, many of the snap-in's supplied enable you to manage other Windows computers on the network as well.
- MMC consoles can be configured to manage other Windows systems by either redirecting the pre-configured console or by creating a custom MMC console and focussing it on the other system

Remote Administration using the MMC

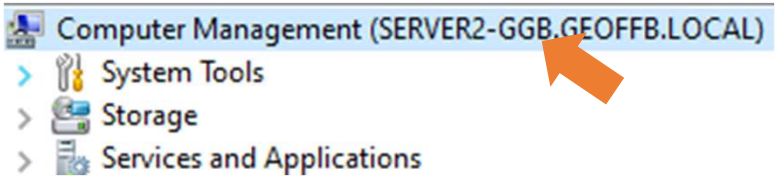
Redirecting existing consoles involves right-clicking the console name and choosing the Connect to another computer option



Do this…

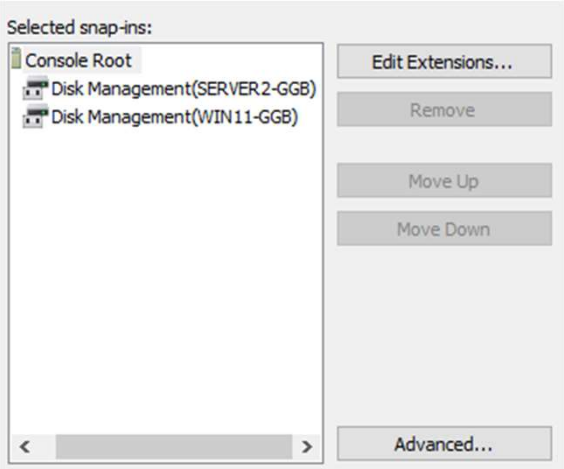


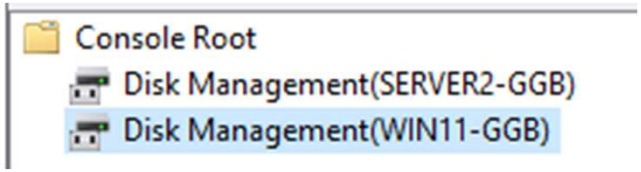
- From Server1 open Computer Management Console
- Right click the root of the console → Connect to another computer.
- Using Browse… query server2



Do this…

- Open MMC on Server1
- Add the disk management snap-in
- In the Select computer dialog query Server2 as the target
- And a second disk management snap-in for win11





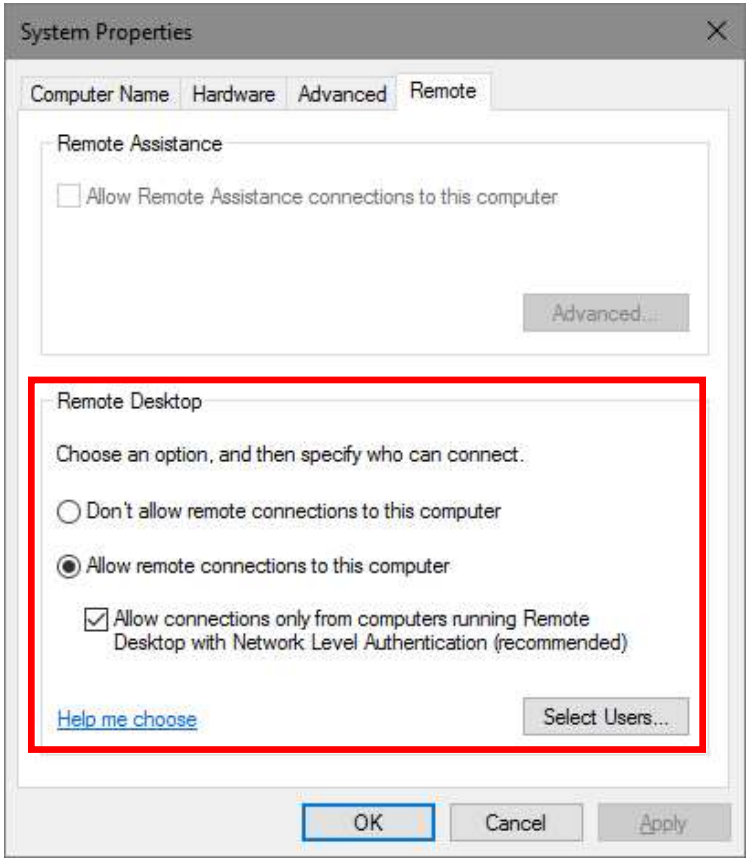
Remote Administration using Remote Desktop

- A full remote-control utility for Windows that requires no interaction from the remote system
- Enables you to operate the remote computer as if you were sitting at the keyboard
- Frequently used to connect to servers that are in secured areas of a network
- Uses a protocol called RDP (Remote Desktop Protocol), on TCP port 3389
- It transmits screen information, keystrokes, and mouse movements between two computers. Any applications you launch in the Remote Desktop client are still running on the remote computer, using its processor and memory resources.

Remote Desktop Restrictions

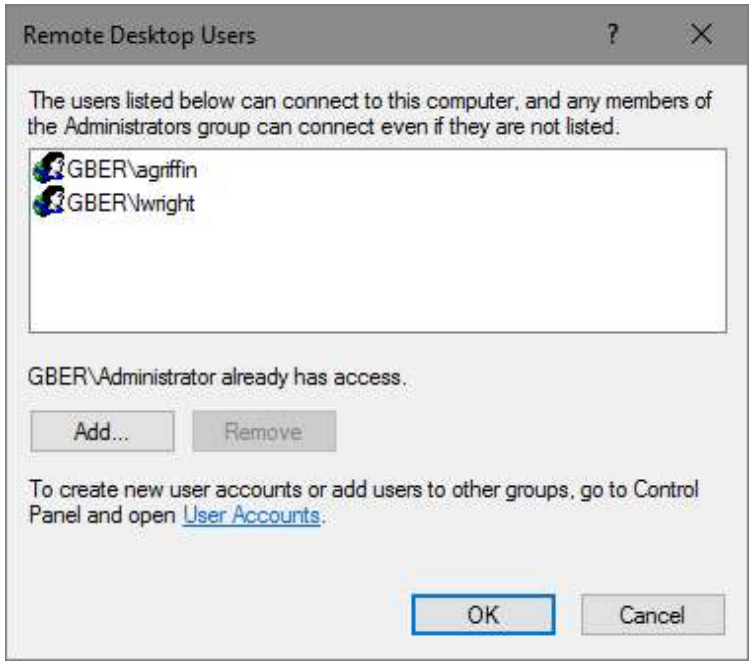
- Windows Server includes a feature known as Remote Desktop for Administration, which allows 2 users to simultaneously connect to the server using Remote Desktop in order to perform server administration tasks.
  - By default, the same user cannot be logged into 2 different simultaneous sessions
- To allow more than 2 users to be simultaneously connected, you need to install Remote Desktop Services Role and purchase Remote Desktop CALs for each user connecting to the server

Enabling Remote Desktop



- Remote Desktop is disabled by default, to enable it, in the Local Server node of Server Manager, click the link beside Remote Desktop and select the Allow remote connections to this computer button

Enabling Remote Desktop



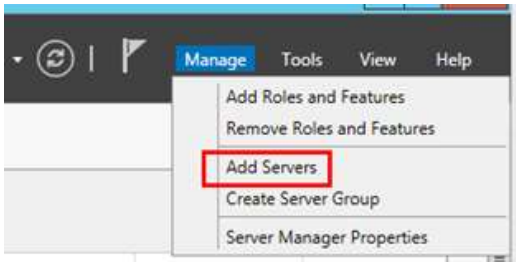
- The Select Users button allows you to specify users who can access a system through remote desktop. (Administrators are granted this privilege by default when remote desktop connections are allowed)
- In addition to being given this permission, users must have the Allow logon locally or Allow Log on Through Remote Desktop Services user right, which is only granted to Administrators on Domain Controllers, by default

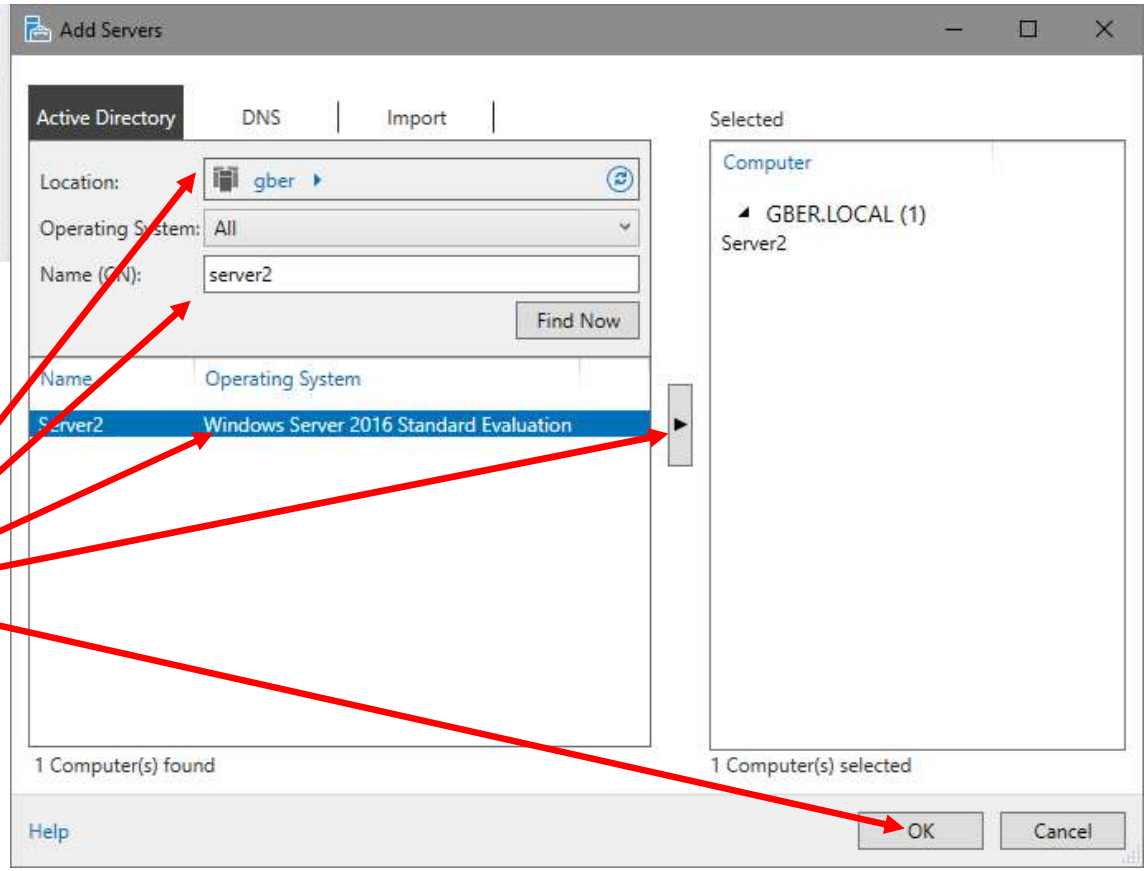
Enabling Remote Desktop (using GPO)

Steps to enable RDP using GPO.

1. Create a new GPO or modify an existing GPO.
2. Enable RDP on target systems using the following setting: Computer Config\Policies\Administrative Templates\Windows Components\Remote Desktop Services\Remote Desktop Session Host\Connections\Allow users to connect remotely using Remote Desktop Services.
   a. Filter the GPO setting on.
3. Computer Config\Policies\Windows Settings\ Security Settings\Restricted Groups\
   a. Add the RDU group
   b. Add the corresponding Access Group to the RDU group in Restricted Groups.
4. Apply the GPO to the OU(s) that contain the computer objects for the target systems.

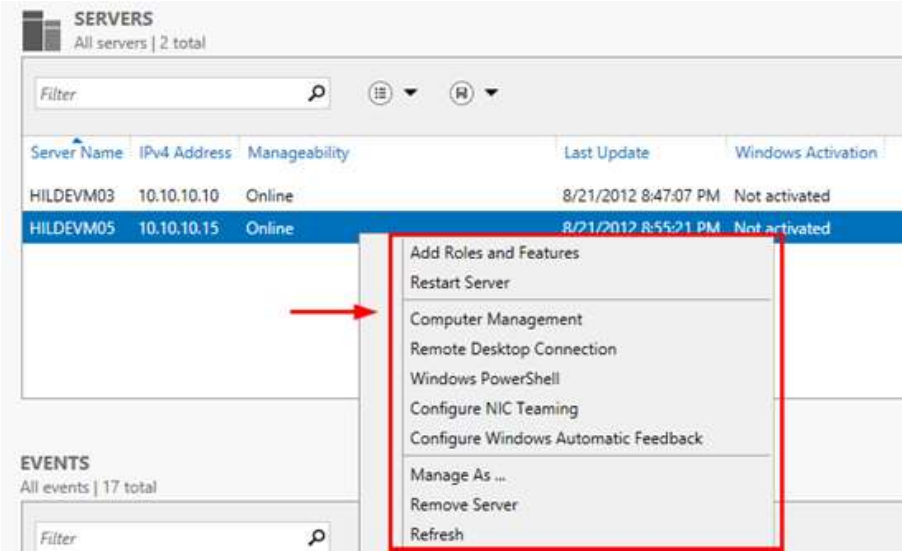
Using Server Manager for Remote Management





Find the servers you are looking for using DNS or import from a file.

Managing Remote Servers with Server Manager

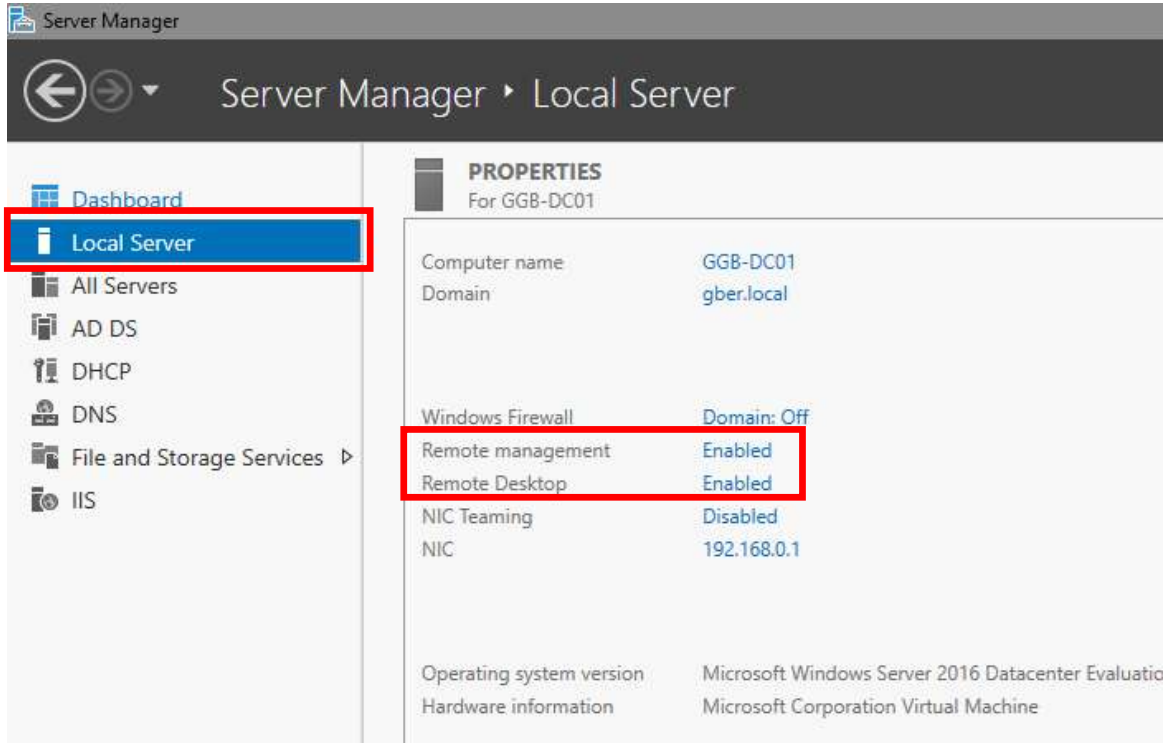


- Once remote servers are added, you can manage them as if they were the local server
- To remove a server from the Pool, right-click it and select 'Remove Server'

Remote Server Administrative Tools (RSAT)

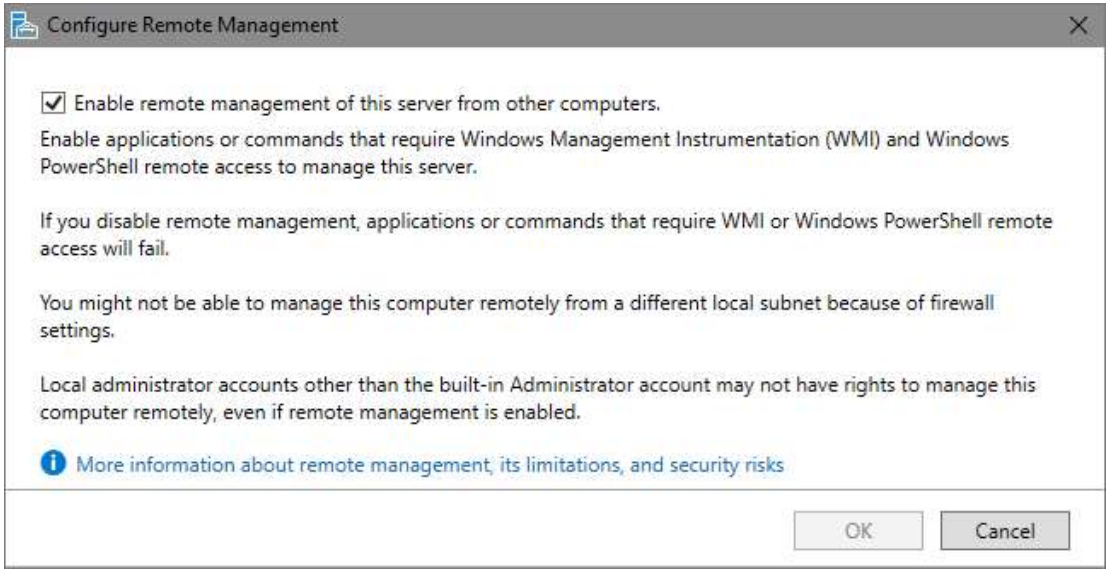
- Microsoft recommends limiting server usage and instead publishes a series of tools called Remote Administration Tools which can be used on Windows client systems to administer Windows Servers remotely.
- Downloaded from Microsoft's website
  https://www.microsoft.com/en-ca/download/details.aspx?id=45520
  - Open an MMC console
  - Add the required snap-in.

Remote Management Status



- On the Local Server homepage, the Properties tile contains a Remote management indicator that specifies the server's current RM status.

Disabling WinRM



- To change the WinRM state, click the Remote Management hyperlink to open the Configure Remote Management dialog box.

Configuring Windows Firewall for Remote Management

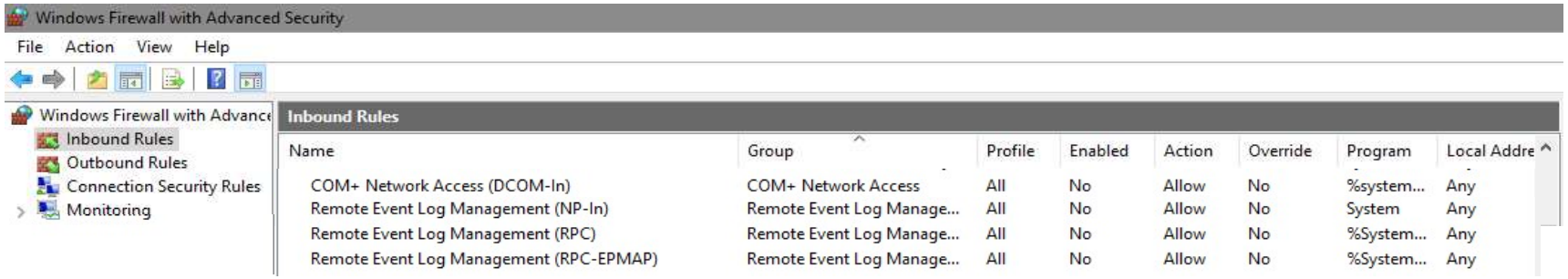
- Enabling WinRM on a server provides access to a number of configuration utilities, however, if you want to enable MMC management of another server, you must make firewall changes to allow the protocols necessary for the MMC to function

Windows Remote Management (HTTP-In)	Windows Remote Manage...	Domai...	Yes
Windows Remote Management (HTTP-In)	Windows Remote Manage...	Public	No

(or turn the firewall off)

Enabling Firewall Rules to Allow MMC Management

- Use the Windows Firewall with Advanced Security console to enable the following rules (wf.msc):
  - COM+ Network Access (DCOM-in)
  - COM+ Remote Administration (DCOM-In)
  - Remote Event Log Management (NP-In)
  - Remove Event Log Management (RPC)
  - Remote Event Log Management (RPC-EPMAP)



Managing Windows Firewall with PWSH

- To use MMC remotely (among other things) you must either enable the appropriate rules in WF or turn the firewall off.
- To Disable/Enable the Firewall:

```
Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False
Get-NetFirewallProfile -Profile Domain,Public,Private
```

- To check the state of a specific FW rule (using RDP as an example):

```
get-Netfirewallrule -DisplayGroup "Remote Desktop" |
     format-table Name, Enabled -autosize
```

Active Directory Administrative Center

Active Directory Administration Center

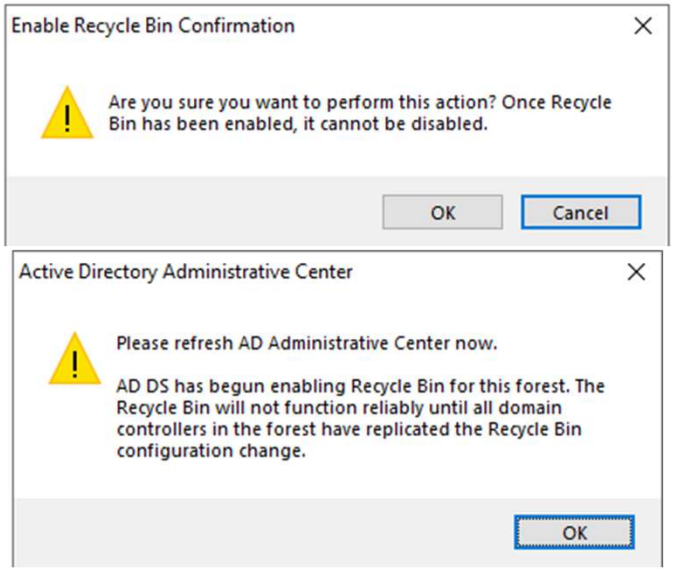
- Introduced in Server 2008
- Administrative Center (AD AC) in Windows Server includes enhanced management experience features.
  - Active Directory Recycle Bin
  - Fine-Grained Password Policy
  - Windows Powershell history

AD Recycle Bin Concepts

- Tombstoning
  - Object in AD is deleted (originating delete)
  - Renamed and stored in the Deleted Objects OU
  - Object is marked as a “tombstone” for 60 days
  - After 60 days, the object is deleted permanently.

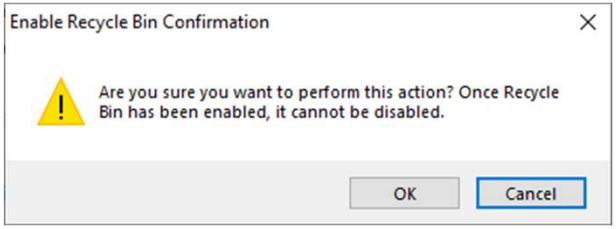
AD Recycle Bin

- By default, Active Directory Recycle Bin is disabled.
- To enable it, raise the forest functional level of your AD DS to **Windows Server 2008 R2** or **higher**.
  - This requires that all domain controllers in the forest or all servers that host instances of AD LDS configuration sets be running Windows Server 2008 R2 or higher.
- The process of enabling Active Directory Recycle Bin is *irreversible*.
- To manage the Recycle Bin feature through a user interface, you must install the version of Active Directory Administrative Center



Enabling and Using AD Recycle Bin

1. Ensure the FFL and DFL are at least Server 2008 or higher.
2. Enable AD RB
   1. Open AD AC
   2. Select your Domain
   3. (Tasks on the right) Enable Recycle Bin



3. Once enabled, deleted items will appear in the Deleted Objects OU.
   1. In AD AC, you can select them, and retore them to their original location, or a new location
4. You can *Reanimate* a deleted object in AD AC
   1. Go to the Deleted Objects OU
   2. R-Click the Object → Restore or Restore to…

Do this…

1. Create and OU called RecycleMe.
2. Enable AD RB in AD AC.
3. Delete RecycleMe
4. Show RecycleMe in the Deleted Items container in AD AC.
5. Restore Delete me to its original location.

Fine-Grained Password Policy

- Using GPO, only one password policy and account lockout policy could be applied to all users in the domain.
- To define different password and account lockout policies for different sets of users in a domain, we use the Fine-Grained Password policy.
  - **Can only be configured in AD AC**.
  - Must be Server 2012 or **greater**.
  - Can only be applied to Global Security Groups and user objects.
  - FFL/DFL must be 2008 or greater
- You can use fine-grained password policies to specify multiple password policies within a single domain and apply different restrictions for password and account lockout policies to different sets of users in a domain.

Fine-Grained Password Policy

- Fine-grained password policies apply only to global security groups and user objects
- Each FGPP has a precedence setting. If more than on FGPP applies to an object, the lower precedence setting applies (wins).

Setting a FGPP in AD AC.

1. Open AD AC → Domain Root → System → Password Settings Container
2. R-Click → New.
    1. Configure Password settings and lockout settings
    2. Add a Global Sec. Group or a User object to the “Directly Applies to” section.
3. Click ok.

Do this…

1. Create a Fine-grained password policy for smattis with the following settings
   1. Precedence = 1
   2. MinPWlength = 4
   3. Enforce PW history = OFF
   4. Minimum age = OFF
   5. Max pass age = 365 days
   6. Failed attempts = 12
   7. Reset after = 1
   8. Lockout duration = 5
   9. Applies Directly to = Smattis

Monitoring System Performance

Monitoring Methodology

- Monitoring is the main method in which administrators maintain the system's integrity over time.
- Monitoring consists of
  - Examining network activity
  - Maintaining and viewing log files
  - Performance testing
  - Problem analysis.
- Monitoring also requires both proactive and reactive maintenance.

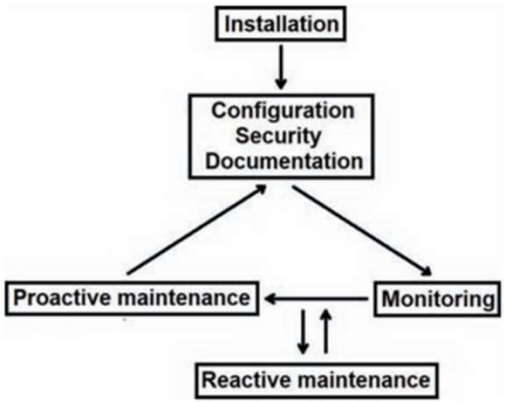


Figure 12-1 The monitoring and maintenance cycle

Monitoring Methodology

- Proactive Maintenance: taking the required steps to minimize the chance of future problems.
  - Performing regular system backups
  - Monitoring available storage
  - Creating baselines
  - Analyzing performance trends to identify problems before they occur
  - Documentation

Monitoring Methodology

- Reactive maintenance: correcting problems when they arise during the monitoring cycle.
  - Solving issues (failures, bottlenecks, misconfigurations) as they occur
  - Documenting the problem and solution with the goal of either preventing future problems and/or decreasing the amount of time the problem affects the environment should it reoccur.
  - Adjusted proactive maintenance accordingly to prevent future failures.

Fixing the Problem

- Collect information:
  - Read log files, examine deviance from baseline performance indicators, live testing.
- Isolate the problem:
  - Determine which subsystem has failed, on which computer(s).
  - Determine the breadth of the problem.
  - Prevent further propagation of the issue.
- Implement a solution:
  - Prioritize the problems according to severity
  - Address the root of the problem, and not simply the effects.
  - If the effects are severe, consider an interim solution.
- Tests the Solution:
  - Possibly in a test environment or on a small subset of the network.
  - Confirm the solution works as intended and does not create new problems
  - Implement the Solution
- Documentation:
  - The goal here is preventing future problems and/or decreasing the amount of time the problem affects the environment should it reoccur.

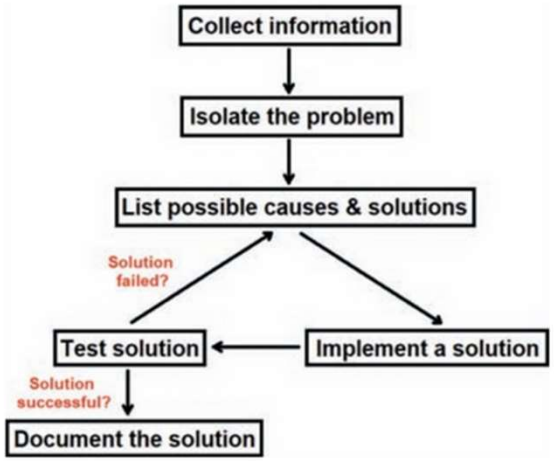
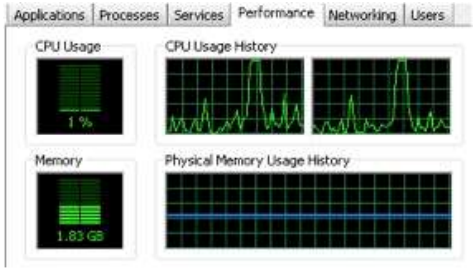


Figure 12-2 Common troubleshooting procedures

Monitoring System Performance

- Two basic types of system monitoring:
  - Real-Time Monitoring
    - Display a continuous stream of current performance statistic
  - Logged Monitoring
    - performance statistics are recorded on a permanent medium instead of simply displaying it.
    - enables administrators to observe trends over time
    - captured data must be reviewed periodically



- When monitoring system performance, the most important element to monitor is the ***change*** (delta) in performance values over time.

Monitoring Tools

- Task Manager (real-time)
- Resource Monitor (real-time)
- Performance Monitor (real-time/logged)
- Data Collector Sets (logged)
- Event Viewer (logged)
- Reliability Monitor (real-time/logged)

Baseline

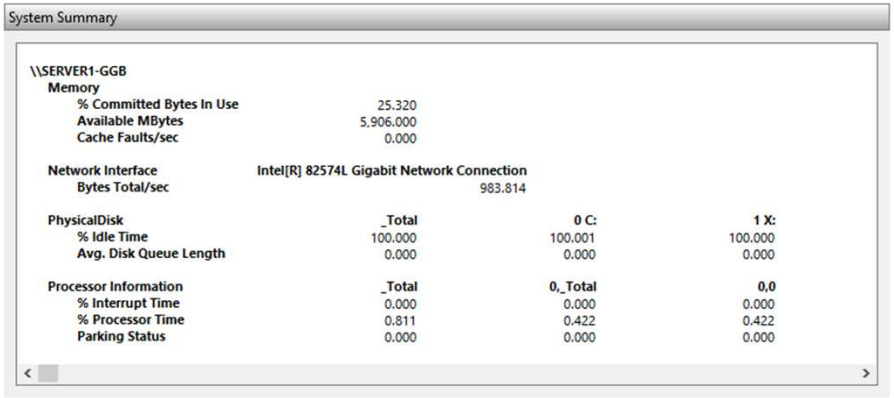
- A baseline is a set of performance measurements taken during periods of normal system activity.
- When performance issues arise, you can compare current performance measurements to those in the baseline.
- Values that have changed dramatically from the baseline can help you pinpoint the source of the performance problem

Bottleneck

- A component that is not providing an acceptable level of performance compared to the other components in the system
- Bottlenecks can appear for a variety of reasons, including:
  - Increased server load
  - Hardware failures
  - Changed server roles

Performance Monitor (console)

- The Performance Monitor console contains
  - Performance Monitor (application) – real time monitoring graph.
  - Data collector sets – (a way of permanently capturing performance data)
  - Reports – the results of our logged monitoring.
- The console root shows a summary of command performance counters, at high level.
  - It shows the four major subsystems: Memory, Network, Disk, Processor

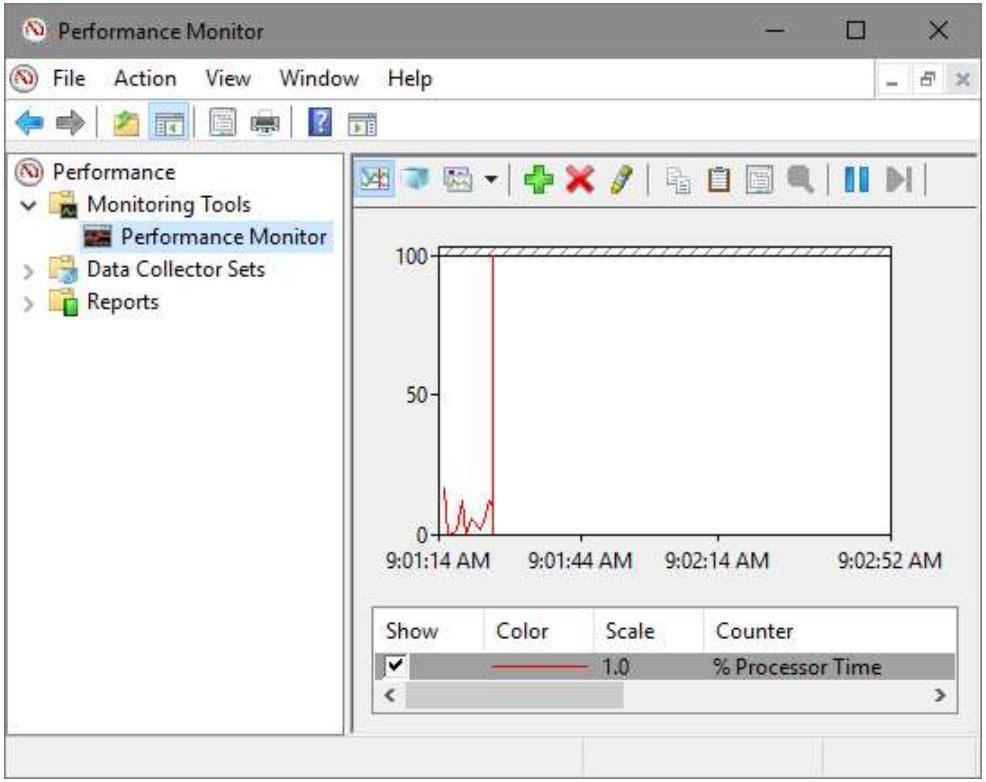


System Summary

\\SERVER1-GGB			
Memory			
% Committed Bytes In Use	25.320		
Available MBytes	5,906.000		
Cache Faults/sec	0.000		
Network Interface	Intel[R] 82574L Gigabit Network Connection		
Bytes Total/sec	983.814		
PhysicalDisk	_Total	0 C:	1 X:
% Idle Time	100.000	100.001	100.000
Avg. Disk Queue Length	0.000	0.000	0.000
Processor Information	_Total	0,_Total	0,0
% Interrupt Time	0.000	0.000	0.000
% Processor Time	0.811	0.422	0.422
Parking Status	0.000	0.000	0.000

Performance Monitor (application)

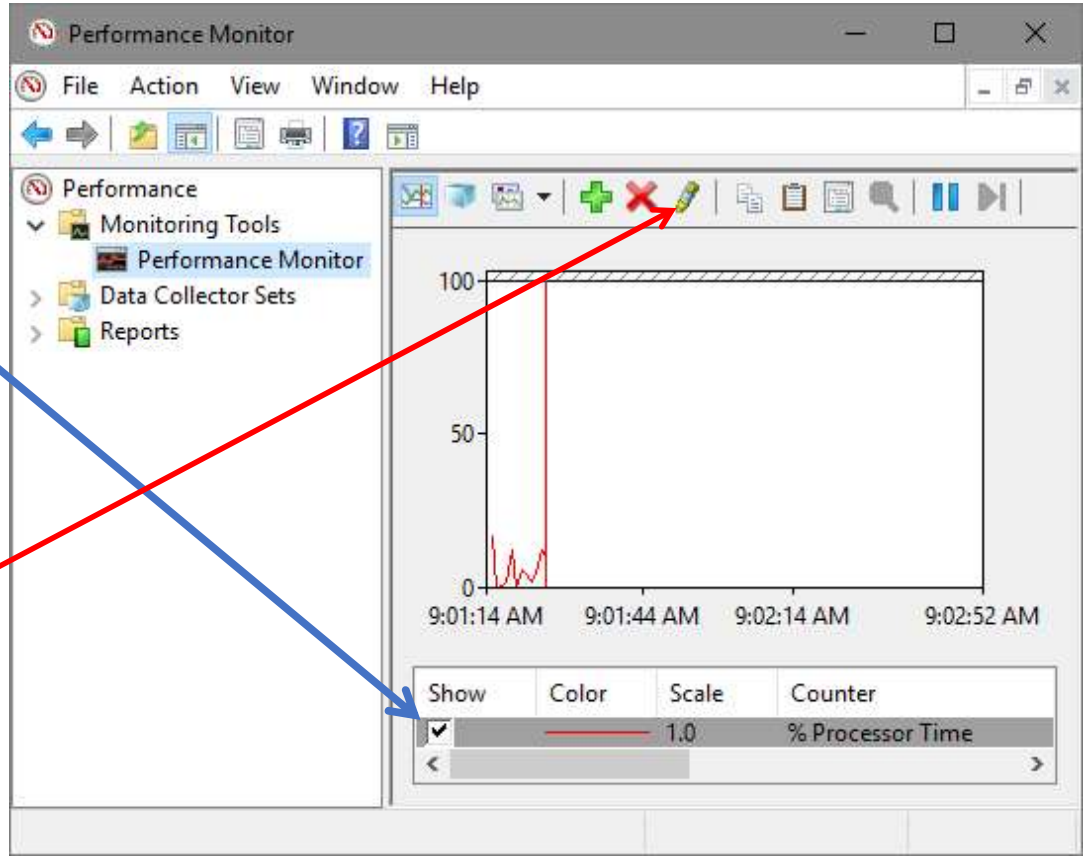
- Displays real-time performance data as collected from components called performance counters.
- By default, it displays a line graph with one performance counter loaded (Processor Time).



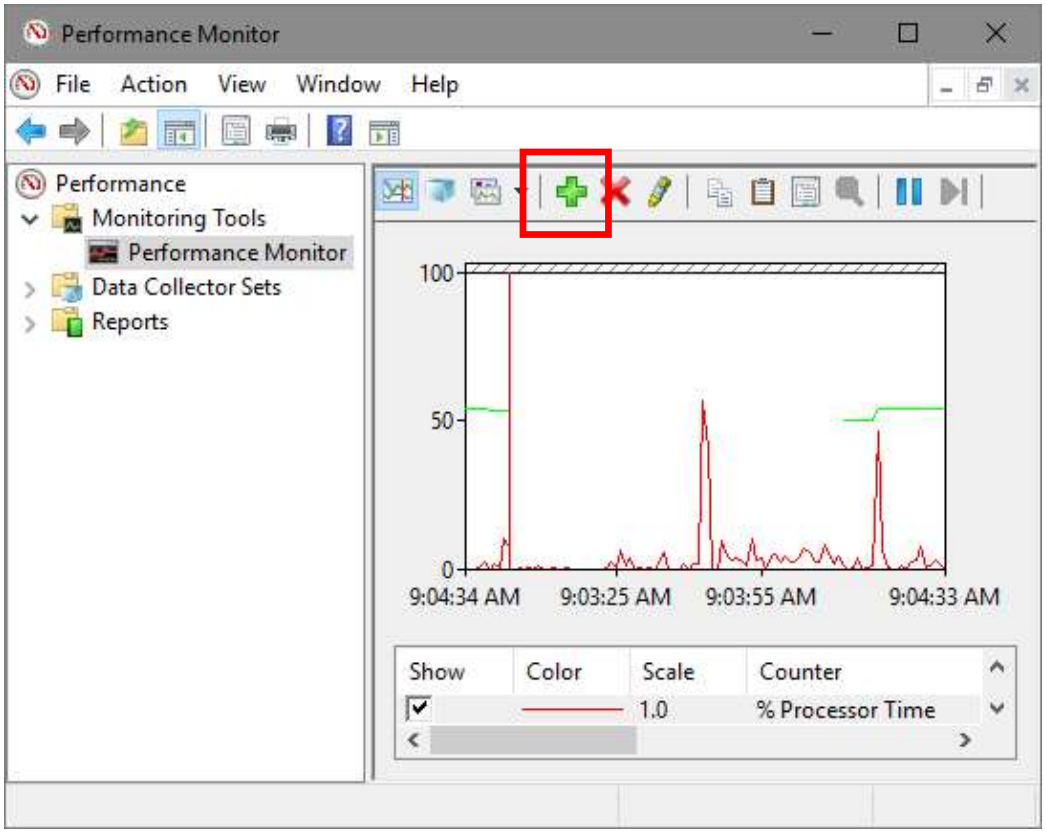
Performance Monitor

- Underneath the line graph, all performance counters are listed as well as their current numerical value.

- When selecting a counter in the lower section select the Highlight button (*looks like a highlighter*) to highlight the line in the line graph that corresponds to the counter.



Performance Monitor



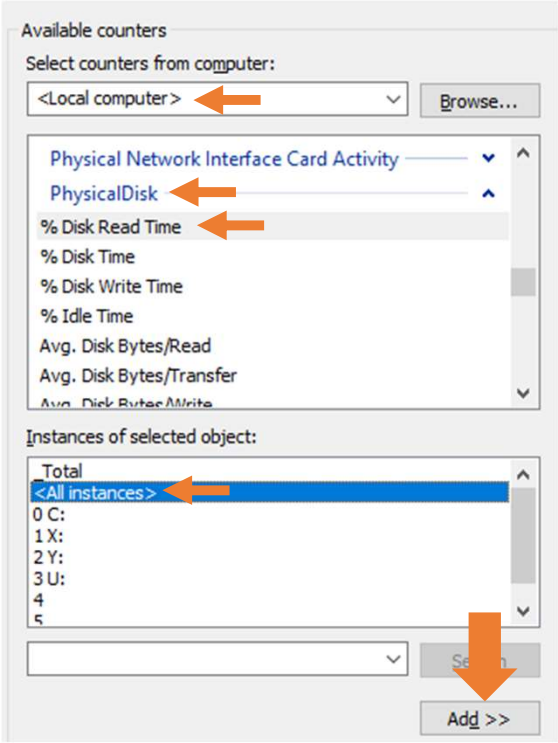
- By selecting the Add Counter toolbar button, you can add different performance counters to the System Monitor console

Performance Monitor Counters

- Four pieces of information must be provided to add additional counters:
  - Computer
    - the computer to monitor
  - Performance Object
    - specific hardware or software component to monitor
    - For example: Processor, Memory, Physical Disk
  - Performance Counter
    - A specific aspect of the performance object's activities
    - For example: %Processor Time, Pages/Sec
  - Instance
    - an element representing a specific occurrence of the selected performance counter
    - For example, which processor, which hard drive to monitor
- When a performance counter is written, it is usually referenced in one of two forms, either *object:counter* or *object(instance)\counter*.

Do this…

- Add a performance counter for Disk usage:
  - Click the Add button at the top ( )
  - Using the Local computer
  - Scroll down to the Physical Disk section in Available counters.
  - Select %Disk Read Time
  - Under Instances… select <All Instances>
  - Click Add.
  - Click OK.



Performance Monitoring

- All computer have four *major* subsystems: Processor, Memory, Hard Drive and Networking.
- A proper monitoring routine would monitor these four subsystems for changes over time.

Monitoring the Processor Subsystem

- An inadequate or malfunctioning processor array can cause a server to queue incoming client requests, preventing the server from fulfilling them promptly.
- If you suspect the processor subsystem has become the bottleneck for the system:
  - Upgrade the processor to a faster speed
  - Upgrade the processor to a different type (Pentium Celeron to Pentium Core i3)
  - Upgrade the processor to a different architecture (x86 to x64)

Monitoring the Memory Subsystem

- A memory leak is the result of a program allocating memory for use but not freeing up that memory when it is finished using it. Over time, the computer's free memory can be totally consumed, degrading performance and ultimately halting the system.
- If you suspect that memory has become the bottleneck:
  - Increase the amount of physical memory in the system
  - Upgrade the speed of your hard drive (quicker paging)
  - Upgrade the speed/type of your RAM

Monitoring the Disk Subsystem

- Disk wear out over time. Disks are one of the most replaced components of the computer, especially if those disk are platter drives. A drop in throughput may indicate a bad disk.
- If you suspect that the disk subsystem has become the bottleneck:
  - Check your disk health using CHKDSK or comparable 3rd party tools. If your disk are showing errors (back sectors/blocks, write/read errors), replace them immediately).
  - Upgrade your disk subsystem (IDE to SATA, SCSI, Serial Attached SCSI (SAS))
  - Implement RAID configurations (RAID 0)
  - Upgrade software-based RAID to hardware-based RAID
  - Upgrade the speed of your hard drives
  - Increase the number of drives and split frequently accessed data between drives

Monitoring the Network Subsystem

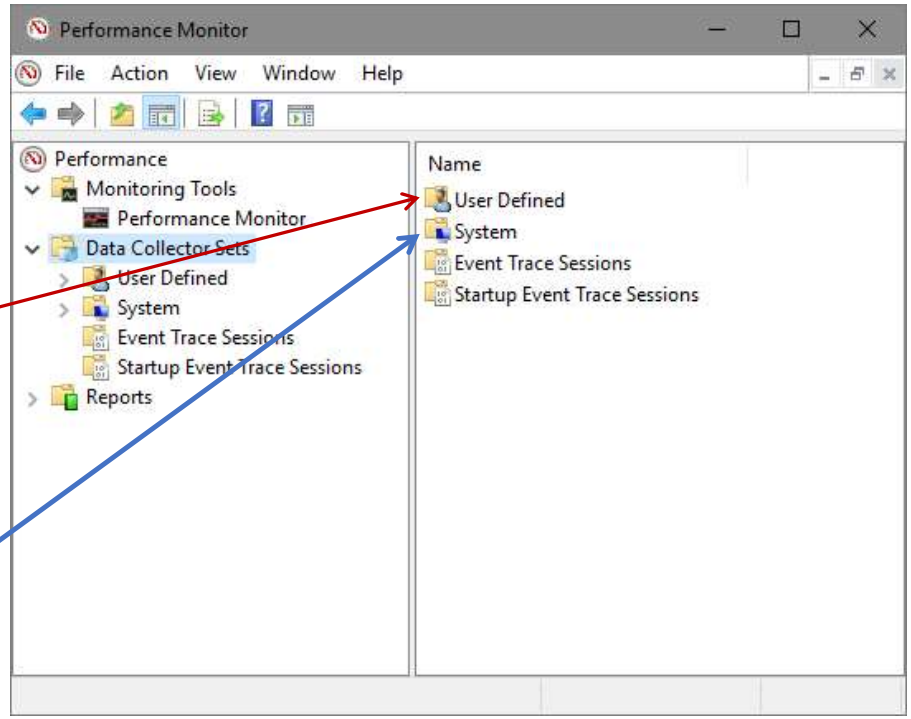
- Networks are complicated to troubleshoot. While most networks are 1GB+ in throughput, there are many factors *not* relating to throughput that can affect performance. Knowing that there *is a bottle neck* probably won't help you determine the issue without further troubleshooting.
- If you suspect that the network subsystem has become the bottleneck:
  - Upgrade the speed of the entire network (100 Mbps Ethernet to 1 Gbps Ethernet)
  - Upgrade Wireless 802.11b or 802.11g networks to 802.11n/ac/ax
  - Install additional network adapters (Nic Teaming)
  - Move network services to another system
  - Troubleshoot layers 1-4 for network errors (duplex mismatches, bad media, broadcast storms, poor Wi-Fi signal strength, policy-based throttling, etc)

Data Collector Sets

- Allow you to specify sets of performance objects and counters that you want to track.
- Data collectors can be used to record performance data on the selected computer at a specific sample interval.

Data Collector Sets

- Under the Data Collector Sets node, you'll find data sets and reports that are user-defined, and system defined.
  - User-defined data sets are created by users for general monitoring and performance turning.
  - System-defined data sets are pre-created by the OS to aid automated diagnostics (the BPA for example).



Creating Data Collector Sets

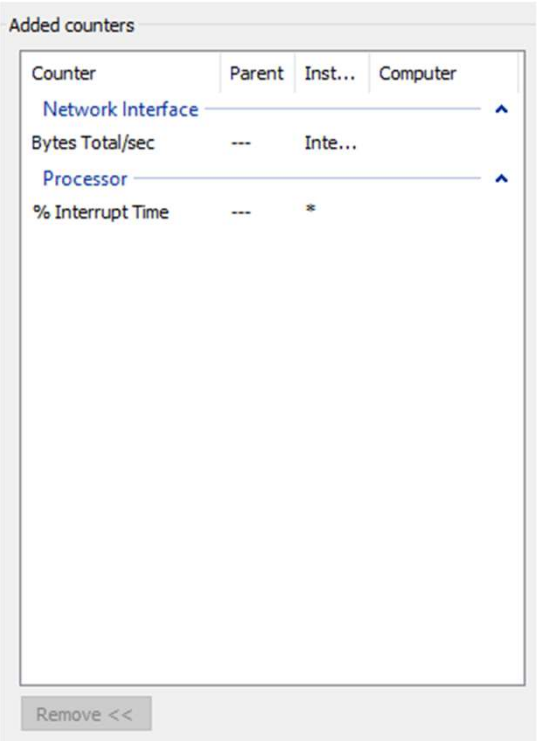
- Right-click the User Defined node in the left pane, point to New and then choose Data Collector Set.
  1. Type a Name.
  2. Select Create Manually (Advanced)
  3. Choose Create Data Logs, Performance Counters
  4. Add counters, specify interval
  5. Select location to store log files
  6. Select credentials to use when launching the log
  7. To schedule the log to start automatically, access the properties of the log and choose the Schedule tab.
  8. Logging will only stop if you set an expiration date for it.

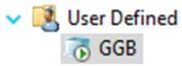
Three types of Data logs.

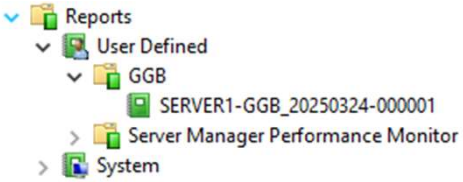
- Performance Counters
  - Monitor various metrics relevant to the CPU, hard disk, Memory and Networking subsystems.
  - Creates histographic datapoints about resource use/consumption.
- Even trace Data
  - Log the occurrence of specific Event logs
- System Configuration information
  - Log the changes in registry keys.

Do this…

1. Create a User Generated Data Collector Set named your initials.
2. Manually configure it.
3. Performance counter.
4. Add
   1. Processor → Interrupts/Sec → All instances
   2. Network Interface → Bytes Total/sec → Intel adapter instance only.
5. Sample interval → 5 sec.
6. Start the Data Collector set (let it run for ~30 seconds). Open a web browser and generate web traffic while you wait.
7. Stop the data collector set.
8. View the report







Viewing Data Collector Sets

- In Performance Monitor, expand the Reports node and then expand the individual report node for the data collector you want to analyze.
- Under the data collector's report node, you'll find individual reports for each logging session.
- A logging session begins when logging starts and ends when logging stops.

Performance Counter Alerts

- Used to notify users when certain events occur or when certain performance thresholds are reached.
- To create a performance counter alert:
  - Right-click the User Defined node in the left pane, point to New and then choose Data Collector Set.
  - Type a name.
  - Select Performance Counter Alert
  - Choose Create Data Logs
  - Add counters
  - Use the Alert When text box to set the occasion when the alert for this counter is to be triggered (above or below a specific value).
  - Select credentials to use when launching the log
  - To schedule the log to start automatically, access the properties of the log and choose the Schedule tab.
- Logging will only stop if you set an expiration date for it.

System Diagnostic

- To have Windows perform a system diagnostic and present the results of that report, use the command `perfmon /report`
  - This will generate a report
  - This will generate a report detailing the status of local hardware resources, system response times, and processes on the local computer along with system information and configuration data.
  - This report includes suggestions for ways to maximize performance and streamline system operation.
- Membership in the local Administrators group, or equivalent, is required to run this report

Reliability Monitor

- Reliability monitor is a graphical way a viewing Event Trends over a period of time.
- Can be used to visually identify reoccurring errors, or to spot transient issues that are triggering events in event viewer.
- Open Reliability Monitor using the `PERFMON /REL` command from CMD, PWSH, or Run.
- The reliability of the system is represented by a blue line. The lower the line, the less reliable a system is.
- Any events that caused a drop in reliability can be viewed in the bottom pane .